Arrow's Theorem: Sen's Proof

We get versions of the proof in two places, first in 3*, then in A1*. We'll do the version in A1*; the one in 3* is trickier.

Overview of proof

The proof runs on a distinction. A group is **locally decisive** over a pair if it gets its way on that pair, and **globally decisive** if it gets its way on every pair.

1. **Spread of Decisiveness.** Local decisiveness over any one pair gives global decisiveness.
2. **Contraction of Decisive Sets.** Take a decisive group with more than one member and split it in two. One of the halves comes out locally decisive over some pair. By 1, that half is globally decisive.
3. **Pareto.** The whole population is decisive.
4. Apply 2 repeatedly, starting from 3. After each application, we get a smaller group that's globally decisive.
5. H is finite, so the sequence stops at a single person, who is globally decisive. That is a dictator.

The setting

A **social welfare function** (SWF) takes a profile of individual orderings $\{R_i\}$, one for each person i in a finite set H, and returns one social ordering R:

$$R = f(\{R_i\})$$

R being an *ordering* means that it is reflexive, complete and transitive. Sen is going to fuss a lot about *transitivity*. The intuitive cases are like the holiday ones we've discussed.

- Holiday in Paris is at least as good as holiday in Hawai'i.
- Holiday in Hawai'i is at least as good as holiday in Paris plus $100.
- But holiday in Paris is not at least as good as holiday in Paris plus $100

Assume at least three distinct alternatives in X, and at least two people.

The four axioms.

- **U (Unrestricted Domain).** f is defined for every logically possible profile of individual orderings.
- **I^2 (Independence of Irrelevant Alternatives).** The social ranking of any pair $\{x, y\}$ depends only on the individual rankings of x and y.
- **P (Pareto Principle).** If everyone prefers x to y, then xPy.
- **D (Non-dictatorship).** There is no person i such that xP_iy implies xPy for every pair, whatever everyone else prefers.

T.A1*.1 (Arrow's Impossibility Theorem). No SWF satisfies U, I^2, P and D together.

Decisiveness

A set $G \subseteq H$ is **locally decisive over the ordered pair (x, y)** iff xPy whenever every member of G prefers x to y, no matter what anyone outside G prefers.

G is **(globally) decisive** iff it is locally decisive over every ordered pair.

A **dictator** is a person i such that $\{i\}$ is globally decisive. So D says no singleton is decisive.

Note that one way to read principle P is that H itself is a decisive group. That will be a key step in any proof.

Step 1: Spread of Decisiveness

T.A1*.2

If G is decisive over any one pair $\{x, y\}$, then G is globally decisive.

Take any other pair $\{a, b\}$, with a, b, x, y all distinct for now. (We aren't entitled to this in general; we're only told there are three options; we'll come back to this.) By U we may consider any profile we like, so consider Table 1:

Table 1: Nobody outside G has been given any ranking of a against b.

	Constraints imposed
Everyone in G	$a \succ x \succ y \succ b$
Everyone outside G	$a \succ x$, and $y \succ b$

Some quick reasoning

1. Everyone prefers a to x, so by **P**, aPx.
2. Everyone prefers y to b, so by **P**, yPb.
3. Everyone in G prefers x to y, and G is locally decisive over (x, y), so xPy.
4. aPx, xPy, yPb. By **transitivity** of the social ordering, aPb. (Note that here we are only using the assumption that P is transitive; an assumption that Sen will mostly *accept*.)

Now apply $\boldsymbol{I^2}$. The social ranking of $\{a, b\}$ depends only on the individual rankings of a and b. Every member of G ranks a above b (by transitivity of their own preference). Nobody outside G was given any ranking of a against b, so the argument above goes through however those rankings are filled in. So for any ranking you like of a and b, provided everyone in G prefers a to b, aPb. So G is locally decisive over (a, b).

The cases where the pairs overlap

The argument above used four distinct alternatives. With only three, we can run the same basic idea with one of a, b identified with one of x, y.

Decisiveness over (x, b). Let G have $x \succ y \succ b$, and let everyone else have $y \succ b$ with x unplaced. Then xPy by decisiveness, yPb by P, so xPb by transitivity. Only G's ranking of $\{x, b\}$ was specified.

Decisiveness over (a, y). Let G have $a \succ x \succ y$, and let everyone else have $a \succ x$ with y unplaced. Then aPx by P, xPy by decisiveness, so aPy by transitivity. Only G's ranking of $\{a, y\}$ was specified.

Every transitivity appeal in this section runs on *strict* preferences, we've only used PP-transitivity so far. The same goes for the step where members of G are said to rank a above b: their own ranking $a \succ x \succ y \succ b$ is all strict, so only individual PP is used there too.

So we can arbitrarily change either the first or second element, and G will still be decisive. But that means for any pair, we can show that G must be decisive.

Step 2: Contraction of Decisive Sets

T.A1*.3.

If G is decisive and has more than one member, then some proper subset of G is decisive.

Partition G into two non-empty pieces, G^1 and G^2. Pick three distinct alternatives x, y, z, and consider this profile:

	Constraints imposed	Left open
Everyone in G^1	$x \succ y$, $x \succ z$	y against z
Everyone in G^2	$x \succ y$, $z \succ y$	x against z
Everyone outside G	none	everything

Everyone in G prefers x to y, and G is decisive, so xPy.

Society ranks z against x somehow, since R is complete. Two cases.

Case A: zRx. With xPy, transitivity gives zPy. Completeness means zRx covers two subcases. If zPx, this is PP again. If zIx, it is IP-transitivity, and that subcase is the only place in the entire proof where anything beyond quasi-transitivity gets used. Which rankings of $\{z, y\}$ were specified? Only G^2's, who all prefer z to y. G^1 and the outsiders were left free on that pair, so the argument holds however their rankings go. By I^2, G^2 is locally decisive over (z, y). By step 1, that shows that G^2 is globally decisive.

Case B: $\sim (zRx)$, i.e. xPz. Which rankings of $\{x, z\}$ were specified? Only G^1's, who all prefer x to z. G^2 and the outsiders were left free on that pair. By I^2, G^1 is locally decisive over (x, z), and by Step 1, G^1 is globally decisive. This case appeals to no transitivity principle at all; xPz follows from $\sim (zRx)$ and completeness alone.

One of the two cases holds, so one of G^1, G^2 is decisive. Both are proper subsets of G.

The result is stronger than the statement (Sen's footnote 7). We did not construct a special proper subset. We took an *arbitrary* partition of G into two pieces and showed one of them must be decisive. Whichever way you cut a decisive set in half, one half is still decisive. In the vocabulary of the end of the chapter, the decisive sets form an ultra-filter.

Step 3: the theorem

1. By **P**, H is decisive.
2. By **T.A1*.3**, some proper subset of H is decisive.
3. That subset, if it has more than one member, has a decisive proper subset. Repeat.
4. H is **finite**, so the sequence terminates. It terminates at a set with one member, who is a dictator, contradicting **D**. ■

How Transitivity is used

Sen's four mixed conditions (Ch 1*, Lemma 1*a).

Label	Condition
PP	$xPy \;\; yPz \to xPz$ (quasi-transitivity)
PI	$xPy \;\; yIz \to xPz$
IP	$xIy \;\; yPz \to xPz$
II	$xIy \;\; yIz \to xIz$

Going through the proof step by step:

Step	What gets chained	Principle used
Spread, step 4	aPx, xPy, yPb	PP, twice
Spread, overlap (x, b)	xPy, yPb	PP
Spread, overlap (a, y)	aPx, xPy	PP
Spread, G ranks a over b	individual preferences	PP, for individuals
Contraction, Case A, zPx subcase	zPx, xPy	PP
Contraction, Case A, zIx subcase	zIx, xPy	IP
Contraction, Case B	nothing	none
Step 3	nothing	none

So the proof uses PP everywhere except one subcase of one case of one lemma, where it uses IP.

There is no way to remove the appeal to PI/IP from that subproof, because we know the theorem doesn't hold with just PP.

- Lemma 1*r(a): if R is complete, PI and IP are equivalent.
- Lemma 1*o(b): if R is complete, PP and PI together imply full transitivity.

R is complete here, so PP together with IP is not a weakening of transitivity. It is transitivity, arrived at from a different direction. The proof never invokes II, and never needs to, because it gets II back from the other two.

What you can prove with quasi-transitivity

Someone who accepts PP and rejects the other transitivity principles still gets a lot of interesting results, e.g.,:

- **Spread of Decisiveness**, in full.
- **Contraction, Case B**, which used no transitivity.
- **Veto-Spread**: if i is semi-decisive over one ordered pair (meaning $xP_iy \to xRy$), then i is semi-decisive over every ordered pair, i.e. has a veto. Sketch: give i the ranking $a \succ x \succ y \succ b$, and everyone else $a \succ x$ and $y \succ b$. P gives aPx and yPb, and semi-decisiveness gives xRy. Suppose bPa. Then yPb, bPa and aPx chain by PP to yPx, contradicting xRy. So aRb.

What you can't prove is Contraction, and with it Step 3. Split Case A the other way, on zPx versus xRz, and the argument stops working:

- if zPx, then PP gives zPy, so G^2 is decisive over (z, y) and hence globally decisive;
- if xRz, we get only that G^1 is *semi*-decisive over (x, z), which by Veto-Spread gives G^1 a veto.

So "some proper subset is decisive" becomes "some proper subset is decisive, or some proper subset has a veto", and that does not shrink to a singleton. It stops at a smallest decisive group each of whose members has a veto, which is Gibbard's oligarchy, T.A1*.4.

A counterexample to make this concrete. The Pareto-extension rule: xPy iff everyone prefers x to y, and xIy otherwise. It satisfies U, I^2, P and D, and yields a reflexive, complete, quasi-transitive R. IP fails: let person 1 rank $x \succ y \succ z$ and person 2 rank $y \succ z \succ x$. Then yPz and xIy, but xIz rather than xPz. And the only decisive set is H itself, so Contraction fails outright. The oligarchy is the whole of society, and every person has a veto.

Where each assumption got used

Each row is a place where someone has tried to argue for keeping all but one part of the Arrow package.

Assumption	Used where	What happens without it
U	Constructing the two profiles	Restrict the domain: single-peakedness (Ch 10)
$\boldsymbol{I^2}$	The last move of each lemma	Allow richer information: cardinality, interpersonal comparison (Ch 7, 9)

Assumption	Used where	What happens without it
P	aPx and yPb; and giving us H decisive at the start	The theorem fails, and so does the appeal of the result
Transitivity	PP throughout Step 1; IP in one subcase of Contraction Case A, and nowhere else	Quasi-transitivity leaves Spread intact and kills Contraction: Gibbard's oligarchy. Acyclicity gives a vetoer
Completeness	Contraction, to guarantee zRx or xPz; and silently, to make PP plus IP amount to full transitivity	Suzumura consistency, and the maximal/best distinction from Ch 1*
$\#X \geq 3$	Both lemmas need a third alternative to route through	With two alternatives, majority rule satisfies everything (May's theorem)
H **finite**	Step 4 only	The contraction never terminates: Kirman and Sondermann's invisible dictators